# Day 5 Task Report

**Title:** Model Behaviour Testing (The Stress Test)

**TASK-05:** Red-Teaming & Systematic Failure Analysis of LLMs

## Objective

The objective of this task is to go beyond normal "happy path" testing and evaluate how a Large Language Model behaves under challenging and adversarial conditions.
This includes identifying where the model fails, produces hallucinations, or does not follow instructions properly.

## Tools Used

- VS Code
- Git & GitHub
- MS Word

## Libraries Used

- **time:** To measure response latency
- **json:** To store and display structured output
- **os & dotenv:** To securely load API key
- **google-genai:** To interact with Gemini API

## Concepts Used

- **Adversarial Testing**: Testing model with tricky or harmful prompts
- **Latency Measurement**: Calculating response time

- **Factuality Check**: Verifying correctness of output
- **Instruction Following**: Checking if model follows given constraints
- **Failure Analysis**: Identifying and documenting model weaknesses

**Technical Implementation**

The system allows the user to input multiple prompts dynamically.
For each prompt:

1. The model generates a response
2. Latency is calculated
3. Factual correctness is evaluated
4. Instruction-following behaviour is checked
5. Results are stored in JSON format

Additionally, a failure report is generated highlighting key failure cases.

**Errors**

**1. API Quota Exceeded (429 Error)**
The API returned a quota exceeded error of specific model.

**Solution:**

Find out all the models that supports that API key and used any one model (gemini-2.5-flash) out of them.

**2. All Outputs Marked as "Pass"**
Even when the model gave incorrect or unsafe responses, the system marked everything as Pass.

**Solution:**

Improved validation logic and Added checks for:

- Safety refusal detection
- Hallucination identification
- Constraint validation

**3. Infinite Loop Confusion**

Program kept asking for input and didn't stop.

**Solution:**

Added clear instruction for user to stop the loop type "exit" and Checked .lower() for proper exit condition.

**Conclusion**

This task provided practical understanding of how Large Language Models behave under stress conditions.
It helped in identifying model limitations such as hallucinations, instruction-following issues, and safety concerns.